

Emanuel Borges

Senior Full Stack Developer · Machine Learning & NLP

Remote (UTC−4) · Employee or contractor · Brazil & international

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PROFESSIONAL SUMMARY

Senior Software Engineer specialized in full stack, mobile and scalable systems architecture. I've worked on large-scale projects in the public and private sectors — INDT, ICCT and Manaus City Hall — focusing on Clean Architecture, testing and DevOps practices. Today I apply Artificial Intelligence and NLP to solve complex problems, especially in semantic search, RAG and information retrieval.

PROFESSIONAL EXPERIENCE

Senior Full Stack Developer

APR 2023 — PRESENT

INDT — Instituto de Desenvolvimento Tecnológico

- Full stack applications with Node.js (NestJS), React, Java Spring Boot and React Native, following Clean Architecture.
- CI/CD pipelines with GitLab, Docker and Kubernetes; quality ensured with SonarQube and code review.
- Scalable solutions with AWS Serverless (Lambda, S3) and messaging with RabbitMQ.

Mid-level Full Stack Developer

JUL 2021 — MAR 2023

ICCT — Instituto CAL-COMP de Tecnologia

- Web systems with Java, PHP/Laravel, Node.js (NestJS), React, Vue.js and Angular.
- Oracle procedures, functions and triggers to optimize queries and business processes.
- Unit testing and Clean Architecture in multidisciplinary agile teams.

Web Developer

AUG 2015 — JUN 2021

Manaus City Hall — SEMEF / SEMAD

- Development of the ComprasManaus and ContratosManaus portals for public procurement and contracts.
- Government systems with PHP, ASP Classic, ASP.NET Web Forms, VB.NET, Java, jQuery, Bootstrap and Oracle.
- Oracle views, functions and procedures for reports and process automation.

SELECTED PROJECTS

Intelligent Legal Critic

RAG · LANGCHAIN · OLLAMA · OWASP LLM

RAG-based legal chatbot that analyzes initial petitions in PDF, suggests improvements, detects prompt injection attempts (OWASP LLM01) and generates Markdown/PDF reports. Orchestrates answers across an internal RAG base, local Ollama and web search.

Risk of Falling Behind in School

SCIKIT-LEARN · XGBOOST · DOCKER

Classification model (Random Forest, XGBoost) with 87% precision for the Passos Mágicos case, with an API and a drift-monitoring dashboard. Top 5% in the datathon.

Stock Forecasting with LSTM

TENSORFLOW · LSTM · FLASK · SWAGGER

LSTM network that forecasts the next stock close with a 1.1–2.2% mean error (MAPE) on an out-of-sample test, evaluated against a naive baseline. Data pipeline via Yahoo Finance and a REST API documented with Swagger.

RAG Agent — Q&A System

LANGCHAIN · CHROMADB · FASTAPI · STREAMLIT

Question-answering app with LangChain, ChromaDB and LLMs (Ollama/Deepseek), with document ingestion, WebSocket chat and continuous feedback.

Financial Hub — P2P Platform

JAVA 17 · SPRING BOOT · KAFKA · REDIS

Hexagonal architecture with PostgreSQL (pessimistic locking), Redis (cache-aside), Kafka with DLQ and MongoDB, plus Angular 19 and Next.js 15 frontends.

Products API

NESTJS · TYPEORM · KAFKA · REDIS

Products API with Redis caching, Kafka events, per-route rate limiting and Swagger docs, on PostgreSQL.

TECHNICAL SKILLS

Backend	Java, Kotlin, Spring Boot, Node.js, NestJS, Express, PHP/Laravel, Python, Flask, FastAPI
Frontend & Mobile	TypeScript, JavaScript, React, React Native, Vue.js, Angular, HTML & CSS
AI & ML	NLP, LLM, RAG, LangChain, TensorFlow, Scikit-learn, Pandas, Hugging Face
Data	PostgreSQL, Oracle, MySQL, MongoDB, SQL Server, Redis
DevOps	Docker, Kubernetes, AWS, Kafka, RabbitMQ, CI/CD, GitLab
Practices	Clean Architecture, Clean Code, TDD, SDD (Spec-Driven Development), Unit testing, Code Review, SonarQube, Scrum

EDUCATION

Postgraduate in Natural Language Processing
Universidade Federal de Goiás (UFG) · 2025 — 2026

Postgraduate in Machine Learning Engineering
FIAP · 2025 — 2026

Postgraduate in Cross-Platform Mobile Apps
Descomplica · 2024 — 2025

Bachelor's in Systems Analysis
FUCAPI · 2000 — 2010

CERTIFICATIONS

Introduction to Node.js — Linux Foundation · Introduction to Cybersecurity — Cisco · Scrum Foundation — Certiprof · Full Cycle: Docker, Kubernetes, GitOps, RabbitMQ · Alura: LangChain with RAG, TypeScript, NestJS, React, CI/CD

LANGUAGES

Portuguese — native · Spanish — intermediate (B2) · English — basic (A2), progressing to intermediate (B1)